Mujoco Paper equations of motion:

$$M(q)dv = (b(q,v) + \tau)dt + J_E(q)^T f_E(q,v,\tau) + J_C(q)^T fc(q,v,\tau)$$

q position in generalized coordinates
v velocity in generalized coordinates
⍰ inertia matrix in generalized coordinates
b "bias" forces: Coriolis, centrifugal, gravity, springs ⍰ external/applied forces ⍰ equality constraints: ⍰ (q)=0
⍰⍰ Jacobian of equality constraints
v∗ ⍰ desired velocity in equality constraint coordinates
f⍰ impulse caused by equality constraints
⍰⍰ Jacobian of active contacts
v⍰ velocity in contact coordinates
f⍰ impulse caused by contacts
⍰ time step 3D rotations (ball joints and base orientations

Dynamics of system: equations of motion
Covered by physics simulator.
Cartpole Lagrangian

This isnt used in Mujoco MPC,
Lagrangian energy method.

$$L(q,\dot{q}) = T(q,\dot{q}) - V(q)$$

where:

T = kinetic energy
V = potential energy

Deriving the Euler Lagrange equations of motion from lagrangian:

It comes from the principle of stationary action.

The derivation goes:

1. define the action
2. perturb the path a little
3. require the first-order change in action to be zero
4. integrate by parts
5. since the perturbation is arbitrary, the Euler–Lagrange equation must hold

1) Start with the action

For generalized coordinates q(t), define

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t)\, dt$$

where L = T - V

Hamilton's principle says the actual motion makes the action stationary:

\delta S = 0

among nearby paths with the same endpoints.

2) Perturb the path

Take a small variation of the trajectory:

$$q_i(t) \rightarrow q_i(t) + \varepsilon \eta_i(t)$$

where:

ε is small

$\eta_i(t)$ is an arbitrary smooth perturbation

endpoints are fixed, so
$\eta_i(t_0) = \eta_i(t_1) = 0$

Then

$\dot{q}_i(t) \to \dot{q}_i(t) + \varepsilon \dot{\eta}_i(t)$

3) Differentiate the action with respect to the variation

$$S(\varepsilon) = \int_{t_0}^{t_1} L(q + \varepsilon\eta, \dot{q} + \varepsilon\dot{\eta}, t)\, dt$$

Stationarity means
$$\left.\frac{dS}{d\varepsilon}\right|_{\varepsilon=0} = 0$$

4) Integrate by parts

$$\int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}_i} \dot{\eta}_i\, dt = \left[\frac{\partial L}{\partial \dot{q}_i} \eta_i\right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) \eta_i\, dt$$

The boundary term vanishes because $\eta_i(t_0) = \eta_i(t_1) = 0$.

$$\delta S = \int_{t_0}^{t_1} \sum_i \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) \eta_i \, dt$$

For stationarity, this must be zero for all admissible \eta_i.
5) Use arbitrariness of the variation

Since each \eta_i(t) is arbitrary in the interior of the interval, the only way the integral is always zero is if the coefficient of each \eta_i is zero:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

This is the Euler–Lagrange equation.

Applying the Euler Lagrange to Cartpole:
We have 1 joint, a robot has multiple joints.

For cartpole, a common choice of generalized coordinates is:

$$q = \begin{bmatrix} x \\ \theta \end{bmatrix}$$

where:

x = cart position
\theta = pole angle

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i$$

Applying the Euler Lagrange equation to a cartpole environment

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i$$

Euler–Lagrange is a general recipe for any mechanical system.
For cartpole, q might be:
$q = [x, \theta]$

For a robot arm, q might be:
$q = [q_1, q_2, \ldots, q_n]$

where each q_i is a joint angle or joint displacement.

The structure is identical, but the expressions inside become more complicated.

Cartpole

Cartpole has just a few coordinates, usually 2:

cart position
pole angle

A small system of coupled equations.

$$q = \begin{bmatrix} x \\ \theta \end{bmatrix}$$

Then you write two Euler–Lagrange equations:

- one for x
- one for \theta

Robot

A robot may have many joints, like 6-DOF, 7-DOF, humanoid, quadruped, etc.

Then:

$$q = \begin{bmatrix} q_1 \\ q_2 \\ \cdots \\ q_n \end{bmatrix}$$

and you get one Euler–Lagrange equation per joint.

In robotics, these are often written in matrix form as:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau + J(q)^T f_{\text{ext}}$$

where:

- M(q) = mass/inertia matrix
- $C(q, \dot{q})\dot{q}$= Coriolis/centrifugal terms
- g(q) = gravity terms

- \tau = actuator torques
- $J^T f_{\text{ext}}$= external contact forces mapped into joint space

That matrix equation is basically the robot version of the Euler–Lagrange equations

Cartpole Example from Lagrangian:
Energy = Kinetic - Potential

- cart position: x
- pole angle: \theta, measured from the upright vertical
- cart mass: m_c
- pole mass: m_p
- distance from pivot to pole center of mass: l
- gravity: g
- horizontal force on cart: u

Generalized coordinates:

q = \begin{bmatrix} x \\ \theta \end{bmatrix}

Generalized forces:

$$Q_x = u, \qquad Q_\theta = 0$$

1) Pole position
Take the pole center of mass to be at

$$x_p = x + l\sin\theta, \qquad y_p = l\cos\theta$$

This choice makes $\theta = 0$ the upright position.

Differentiate:

$$\dot{x}_p = \dot{x} + l\dot{\theta}\cos\theta$$
$$\dot{y}_p = -l\dot{\theta}\sin\theta$$

2) Kinetic Energy

Cart

$$T_{\text{cart}} = \frac{1}{2}m_c\dot{x}^2$$

Pole

For the pole mass center:

$$T_{\text{pole}} = \frac{1}{2}m_p\left(\dot{x}_p^2 + \dot{y}_p^2\right)$$

Substitute velocities:

$$\dot{x}_p^2 + \dot{y}_p^2 = (\dot{x} + l\dot{\theta}\cos\theta)^2 + (-l\dot{\theta}\sin\theta)^2$$

Expand:

$$= \dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta$$

Since $\cos^2\theta + \sin^2\theta = 1$,

$$\dot{x}_p^2 + \dot{y}_p^2 = \dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2$$

So:

$$T_{\text{pole}} = \frac{1}{2}m_p\left(\dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2\right)$$

Total kinetic energy

$$T = T_{\text{cart}} + T_{\text{pole}}$$

$$T = \frac{1}{2}(m_c + m_p)\dot{x}^2 + m_p l\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}m_p l^2\dot{\theta}^2$$

3) Potential energy

Using

$$y_p = l\cos\theta,$$
$$V = m_p g l\cos\theta$$

4) Lagrangian

L= T - V; substitute terms for kinetic energy and potential energy from above

$$L = \frac{1}{2}(m_c + m_p)\dot{x}^2 + m_p l\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}m_p l^2\dot{\theta}^2 - m_p g l\cos\theta$$

5) Euler Lagrange for x

We use

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = Q_x$$

Compute $\partial L / \partial \dot{x}$

$$\frac{\partial L}{\partial \dot{x}} = (m_c + m_p)\dot{x} + m_p l \dot{\theta} \cos\theta$$

Differentiate in time:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = (m_c + m_p)\ddot{x} + m_p l \left(\ddot{\theta} \cos\theta - \dot{\theta}^2 \sin\theta\right)$$

Compute $\partial L / \partial x$

There is no explicit x in L, so

$$\frac{\partial L}{\partial x} = 0$$

Plug into Euler Lagrange

$$(m_c + m_p)\ddot{x} + m_p l \left(\ddot{\theta}\cos\theta - \dot{\theta}^2 \sin\theta\right) = u$$

So the first equation is
$(m_c + m_p)\ddot{x} + m_p l \cos\theta\,\ddot{\theta} - m_p l \sin\theta\,\dot{\theta}^2 = u$

7) Euler–Lagrange for \theta

Now use

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = Q_\theta = 0$$

Compute $\partial L / \partial \dot{\theta}$
$$\frac{\partial L}{\partial \dot{\theta}} = m_p l \dot{x}\cos\theta + m_p l^2 \dot{\theta}$$

Differentiate:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = m_p l \left(\ddot{x}\cos\theta - \dot{x}\dot{\theta}\sin\theta\right) m_p l^2 \ddot{\theta}$$

Compute $\partial L / \partial \theta$

Only two terms depend on \theta:

$$m_p l \dot{x}\dot{\theta}\cos\theta$$

and

$$-m_p g l \cos\theta$$

Differentiate:

$$\frac{\partial L}{\partial \theta} = -\, m_p l \dot{x}\dot{\theta} \sin\theta + m_p g l \sin\theta$$

Apply Euler–Lagrange

$$m_p l \left(\ddot{x}\cos\theta - \dot{x}\dot{\theta}\sin\theta\right) + m_p l^2 \ddot{\theta} - \left(-m_p l \dot{x}\dot{\theta}\sin\theta + m_p g l \sin\theta\right) = 0$$

Distribute the minus:

$$m_p l \left(\ddot{x}\cos\theta - \dot{x}\dot{\theta}\sin\theta\right) + m_p l^2 \ddot{\theta} + m_p l \dot{x}\dot{\theta}\sin\theta - m_p g l \sin\theta = 0$$

The mixed terms cancel:

$$-m_p l \dot{x}\dot{\theta}\sin\theta + m_p l \dot{x}\dot{\theta}\sin\theta = 0$$

So we get

$$m_p l \cos\theta\, \ddot{x} + m_p l^2 \ddot{\theta} - m_p g l \sin\theta = 0$$

Divide by m_p l if desired:

$$\cos\theta\, \ddot{x} + l\ddot{\theta} - g \sin\theta = 0$$

or

$l\ddot{\theta} = g \sin\theta - \cos\theta\, \ddot{x}$

7) Final coupled equations

So the cartpole equations from Euler–Lagrange are:

$(m_c + m_p)\ddot{x} + m_p l \cos\theta\, \ddot{\theta} - m_p l \sin\theta\, \dot{\theta}^2 = u$

$l\ddot{\theta} = g \sin\theta - \cos\theta\, \ddot{x}$

These are the coupled nonlinear equations.

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8) $Solve for \ddot{x} and \ddot{\theta}$

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From the second equation:

$$\ddot{\theta} = \frac{g \sin\theta - \cos\theta\, \ddot{x}}{l}$$

Plug this into the first:

$$(m_c + m_p)\ddot{x} + m_p l \cos\theta \left(\frac{g \sin\theta - \cos\theta\, \ddot{x}}{l} \right) - m_p l \sin\theta\, \dot{\theta}^2 = u$$

Simplify:

$$(m_c + m_p)\ddot{x} + m_p \cos\theta \left(g \sin\theta - \cos\theta\, \ddot{x} \right) - m_p l \sin\theta\, \dot{\theta}^2 = u$$

$$(m_c + m_p)\ddot{x} + m_p g \sin\theta \cos\theta - m_p \cos^2\theta\, \ddot{x} - m_p l \sin\theta\, \dot{\theta}^2 = u$$

Group $\ddot{x}$:

$$\left(m_c + m_p - m_p \cos^2\theta\right)\ddot{x} = u - m_p g \sin\theta\cos\theta + m_p l \sin\theta\,\dot{\theta}^2$$

So

$$\ddot{x} = \frac{u + m_p l \dot{\theta}^2 \sin\theta - m_p g \sin\theta\cos\theta}{m_c + m_p - m_p\cos^2\theta}$$

Then

$$\ddot{\theta} = \frac{g\sin\theta - \cos\theta\,\ddot{x}}{l}$$

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9) Relation to your simulator code

Your code had:

$$\text{temp} = \frac{u + m_p l \dot{\theta}^2 \sin\theta}{m_c + m_p}$$

$$\ddot{\theta} = \frac{g\sin\theta - \cos\theta\ \text{temp}}{l\left(\frac{4}{3} - \frac{m_p\cos^2\theta}{m_c + m_p}\right)}$$

$$\ddot{x} = \text{temp} - \frac{m_p l \ddot{\theta}\cos\theta}{m_c + m_p}$$

That version is very close, but it corresponds to a slightly different physical assumption: it includes the pole's rotational inertia as if the pole were a uniform rod, which is where the \tfrac43 factor comes from.